EVALUATING EMBEDDING MODELS ON DANISH HISTORICAL NEWSPAPERS

A CORPUS AND BENCHMARK RESOURCE

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INTRODUCTION

Historical newspapers

- Mass medium conveying information to a broad public
- Fostering shared discussion within a public sphere, formation of imagined communities
- Offer increases, content diversifies, readership expands

Availability and accessibility

- British, Finnish, Swedish, German, French, and Spanish corpora
- Denmark and Norway: poor quality makes large-scale studies for the analysis of the content difficult or impossible

Goal of this paper

- Make an enriched corpus of Danish historical newspapers accessible for large-scale computational research
- Introduce three custom benchmark tasks to identify the most suitable model for historical Danish texts

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Twenty-eight Danish historical newspapers (1666–1850)

- Printed in the conglomerate state of Denmark-Norway (united until 1814)
- Scans digitally available via Royal Library of Denmark and the National Library of Norway
- Material is set in *fraktur* type, printed on thin, now detoriated, paper sheets with complicated layouts
- Low OCR quality: word-level accuracy $< 50\%$ for the Danish collection

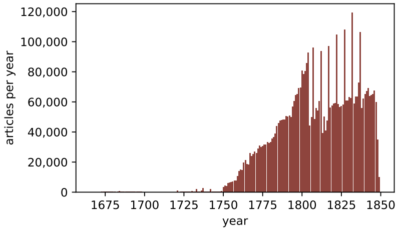

Figure 1: Distribution of the corpus, expressed in number of articles per year.

Newspaper	Period	Editions	Articles	Words
Aalborg Stiftstidende	1827–1846	4,519	203,072	1,9026,204
Aalborgs Stifts Adresseavis	1818–1827	1,591	62,540	5,212,682
Aarhuus Stifts-Tidende	1794–1847	8710	365738	30492484
Adresseavis for Børn	1779–1782	206	1,471	504,229
Almuevennen	1842–1844	161	2,020	708,770
Berlingske Tidende	1749–1836	10,946	792,663	113,926,106
Corsaren	1840–1844	211	2,832	866,368
Danske Mercurius	1666–1691	144	2,308	256,423
Den Nord-Cimbriske Tilskuer	1824–1848	3,238	94,609	9,522,907
Den Vest-Sjællandske Avis	1815–1842	3,621	132,439	16,824,428
Efterretninger fra Adresse-Contoiret i Bergen	1765–1814	2,175	86,338	5,740,796
Extraordinaire Maanedlige Relationer	1672–1698	250	7,372	876,360
Extraordinaire Relationer	1720–1748	338	6,887	714,399
Jyllandsposten	1838–1841	250	3,471	933,957
Jyske Efterretninger	1767–1823	4,485	150,227	14,433,761
Kiøbenhavns Extraordinaire Relation	1721–1745	341	2,796	485,171
Kiøbenhavns Maanetlige Postrytter	1731–1748	213	4,172	470,029
Kiøbenhavns Postrytter	1733–1798	1,920	46,860	4,268,815
Københavns Adresseavis	1759–1837	11,425	1,377,081	105,900,429
Lolland-Falsters Stifts-Tidende	1809–1847	4,435	163,765	14,303,036
Norske Intelligenssedler	1763–1814	2,957	175,327	12,426,028
Nye Tidender	1698–1730	54	891	111,246
Nyeste Skilderie af Kiøbenhavn	1803–1831	2,891	73,059	13,124,176
Odense Adresse-Contoirs Efterretninger	1772–1848	12,155	520,007	45,674,584
Politivennen*	1798–1809	-	4,051	1,071,413
Ribe Stifts Adresseaviser	1786–1848	6,132	238,260	20,565,322
Tronhiems Adresse-Contoirs Efterretninger	1767–1799	1,683	33,808	3,012,461
Viborger Samler	1773–1849	8,845	344,020	31,684,297
Total		93,908	4,898,084	473,136,881

CORPUS

Three subgenres

- Early newspapers (<1800): European style, largely focusing on royal and diplomatic news
- Local/National newspapers (late 18th century): mixed, reaching broader reading public
- Opinion-based papers (from 19th century)

Group	**Period**	**Editions**	**Articles**	**Words**
Early newspapers	1666–1798	3,261	71,186	6,972,443
Local/National papers*	1749–1848	77,778	3,941,251	436,083,783
Opinion-based papers	1798–1844	12,662	124,176	29,576,426
Outlier: *Adresseavis for Børn***	1779–1782	206	1,471	504,229
Total		93,908	4,898,084	473,136,881

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1. Introduction
2. Corpus
3. **Methods**
   - OCR processing the corpus
   - Benchmark task I
   - Benchmark task II
   - Benchmark task III
4. Results and discussion
5. Conclusion and future work

OCR PROCESSING THE CORPUS

Re-digitization via Transkribus

- Custom layout and text recognition models trained on the collections
- Main challenge: separating columns with hairline vertical separators
- Built on ~420,000 manually transcribed words (1750–1850), using Transkribus' implementation of PyLaia framework

Performance and design choices

- Character error rate: **0.56%** → word-level accuracy **96–97%**
- Diplomatic transcription → no spelling standardisation, no post-OCR correction
- Articles segmented via line-level classification (`SetFit` + `RandomForest`)
- Per-text predicted word accuracy score assigned; corpus median: **95.8%**

MODELS TO BE TESTED

Model	Language	Max tokens	Dimensions	Layers	Derived from
`MeMo-BERT-03`	Historical Danish	514	768	12	DanskBERT
`Old_News_Segmentation_SBERT_V0.1`	Historical Danish	512	768	12	DA-Bert_Old_News_V1
`bge-m3`	Multilingual	8,192	1,024	24	XLM-RoBERTa-large
`embeddinggemma-300m`	Multilingual	2,048	768*	24	T5Gemma
`jina-embeddings-v3`	Multilingual	8,192	1,024*	24	XLM-RoBERTa-large**
`multilingual-e5-large`	Multilingual	514	1,024	24	XLM-RoBERTa-large

BENCHMARK TASK I: ARTICLE CATEGORY CLASSIFICATION

Gold sample

- Manually annotated balanced sample of 1,500 articles, taken from 11 local/national newspapers
- Categories: *National news*, *International news*, and *Advertisement*

Experiments with three categories

- `LogisticRegression` model trained in 50 iterations (80/20 train/test split)
- TF/IDF as baseline experiment $(ngram\_range = (1, 1), max\_features = 10,000)$
- Semantic embeddings of the selected six models as features

Experiments with four categories

- Added 180 *Miscellaneous* articles (paratext such as headers) to gold sample to compare performances

Model	Precision			Recall			F1-score		
	ads.	int.	nat.	ads.	int.	nat.	ads.	int.	nat.
TF/IDF	0.937	0.885	0.853	0.926	0.952	0.878	0.931	0.917	0.865
`MeMo-BERT-03`	0.939	0.951	0.902	0.937	0.956	0.897	0.937	0.954	0.899
`Old_News`	0.969	0.972	0.938	0.958	0.974	0.944	0.963	0.973	0.941
`e5`	0.945	0.926	0.890	0.927	0.957	0.874	0.935	0.941	0.881
`jina`	0.926	0.936	0.870	0.915	0.939	0.875	0.920	0.937	0.872
`bge-m3`	0.949	0.951	0.884	0.917	0.956	0.907	0.932	0.953	0.895
`gemma`	0.832	0.845	0.775	0.821	0.879	0.750	0.825	0.861	0.762

Table 1: Performance metrics for the three-class classification task using `LogisticRegression` models with different embeddings as features and TF/IDF representations as baseline.

Model	Precision				Recall				F1-score			
	ads.	int.	nat.	misc.	ads.	int.	nat.	misc.	ads.	int.	nat.	misc.
TF/IDF	0.937	0.885	0.853	0.996	0.926	0.952	0.878	0.663	0.931	0.917	0.865	0.792
`MeMo-BERT-03`	0.933	0.950	0.889	0.928	0.931	0.955	0.896	0.884	0.931	0.952	0.892	0.903
`Old_News`	0.958	0.971	0.930	0.933	0.955	0.974	0.936	0.901	0.956	0.972	0.932	0.915
`e5`	0.915	0.907	0.819	0.968	0.922	0.960	0.873	0.508	0.918	0.932	0.845	0.660
`jina`	0.902	0.915	0.823	0.957	0.906	0.938	0.875	0.641	0.904	0.926	0.848	0.765
`bge-m3`	0.940	0.940	0.858	0.960	0.906	0.960	0.915	0.787	0.922	0.950	0.885	0.862
`gemma`	0.810	0.837	0.758	0.840	0.812	0.880	0.757	0.675	0.811	0.857	0.757	0.746

Table 2: Performance metrics for the four-class classification task using `LogisticRegression` models with different embeddings as features and TF/IDf representations as baseline.

BENCHMARK TASK II: FICTION/NON-FICTION CLASSIFICATION

Gold sample

- Focusing on serialized fiction fragments published in newspapers
- Manually annotated balanced sample of *fiction* $(n = 821)$ and *non-fiction* $(n = 821)$ fragments
- Assigning an ID to each serialized piece of the same series

Classification experiment

- `StratifiedGroupKFold` to avoid pieces from same series in train and test
- `LogisticRegression` model trained with 5-fold cross-validation
- TF/IDF as baseline experiment $(ngram\_range = (1, 1), max\_features = 10,000)$
- Semantic embeddings of the selected six models as features

Model	Precision		Recall		F1-score		Accuracy
	fiction	non-fiction	fiction	non-fiction	fiction	non-fiction	
TF/IDF	0.889	0.837	0.820	0.904	0.853	0.869	0.863
MeMo-BERT-03	0.8749	0.882	0.879	0.880	0.878	0.880	0.880
Old_News	0.890	0.883	0.881	0.891	0.885	0.887	0.886
e5	0.884	0.871	0.867	0.887	0.874	0.878	0.876
jina	0.863	0.868	0.869	0.862	0.865	0.865	0.865
bge-m3	0.865	0.871	0.870	0.866	0.867	0.868	0.867
gemma	0.827	0.817	0.816	0.830	0.821	0.823	0.822

Table 3: Performance (averaged across 5 folds) of TF/IDF and six embedding models on the fiction/non-fiction classification task, evaluated using 5-fold `StratifiedGroupKFold` cross-validation with group-preserving splits. Metrics reported are precision, recall, and F1-score (per class). Note that the second-best model (`e5`) sometimes has a higher precision or recall for one of the classes, while `Old_News` performs more consistently (i.e., has slightly higher F1 for fiction).

BENCHMARK TASK III: FEUILLETON CLUSTERING

Gold sample

- Consisting of *fiction* and *non-fiction* (only those with IDs, i.e. those continuing over multiple fragments)

Clustering experiment

- `K-means` clustering, number of clusters $==$ number of unique IDs
- TF/IDF as baseline experiment $(ngram\_range = (1,1), max\_features = 10,000)$
- Semantic embeddings of the selected six models as features

Model	ARI	V
TF/IDF	0.078	0.618
`MeMo-BERT-03`	0.100	0.613
`Old_News`	0.086	0.659
`e5`	0.114	0.618
`jina`	0.262	0.785
`bge-m3`	0.122	0.676
`gemma`	0.034	0.055

Table 4: Clustering performance of different embedding models on fragment of the same story or parts of series. We only used the datapoints from the classification task that had a continuation, i.e., 163 series in total. Rows are colored based on their performance (V score).

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BENCHMARK TASK PERFORMANCE

Classification (Benchmark tasks I & II)

- `Old_News` outperforms all models in both article category and feuilleton classification
- TF/IDF remains a surprisingly strong baseline
- `MeMo-BERT-03` (trained on ~850 novels) only matches `bge-m3`, likely due to limited training data
- `gemma` underperforms consistently across all tasks

Clustering (Benchmark task III)

- `jina` outperforms all others; `bge-m3` is runner-up
- Larger context window (8,192 tokens) aids cross-sentence clustering
- Higher embedding dimensionality (1,024 vs. 768) likely encodes richer semantic detail
- `jina`, `bge-m3`, and `e5` all share an XLM-RoBERTa-large backbone — the top-3 ARI scorers

MODEL CHOICE AND ITS IMPLICATIONS

Key takeaway: No single model wins across all tasks

Factors favouring `Old_News`

- Domain & temporal specificity
- Strong classification performance
- Locally hostable — lower cost & environmental footprint

Factors favouring `jina/bge-m3`

- Longer context window
- Higher-dimensional embeddings
- `XLM-RoBERTa-large` backbone
- Superior for clustering tasks

CORPUS ENRICHMENT

`Old_News` **as selected embedding model**

- Most reliable and interpretable across tasks

Predict article categories across full corpus

- Confirms distributional differences between newspaper subgenres

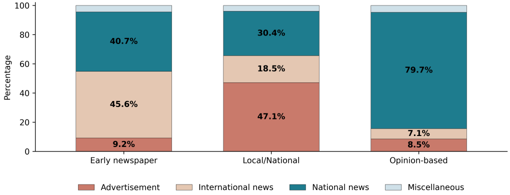

Figure 2: Average category distribution by newspaper groups in the corpus.

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LIMITATIONS

OCR Quality

- Newspapers printed in *fraktur* on thin paper, later microfilmed → uneven type, worn paper, and scanning noise affect transcription quality.

Article category labels

- Categories are *predicted*, not gold-standard
- Four-class scheme forces all articles (incl. *fiction*) into predefined categories

Model specifications

- Context window, embedding dimensionality, and backbone architecture interact in complex ways

CONCLUSIONS & FUTURE WORK

What we did

- Introduced an enriched dataset of Danish historical newspapers (late 17th–19th c.), ~5 million articles with semantic embeddings
- Benchmarked six embedding models across three tasks (classification $\times$ 2, clustering $\times$ 1)
- Released corpus and `Old_News` embeddings openly via HuggingFace

Future work

- Geographical horizons of reporting and information flow (*National/International news*)
- Targeted analysis of fictional content in the press
- Broader computational historical linguistics and digital cultural heritage research

THANK YOU

`try: questions?`

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Code: `https://github.com/centre-for-humanities-computing/danish-newspaper-embs`
Corpus and embeddings: `https://huggingface.co/datasets/chcaa/eno-embs-old-news`